

Data (given)

$CI_{r,t}$
carbon intensity

$CFE_{r,t}$
carbon-free
fraction

D_t^{flex}
flexible demand

$\mathcal{R}, \mathcal{T}$
regions, hours

Parameters (set)

$\alpha, \sigma, \delta, \kappa, \rho, \eta$
parameters

Linear program

Decision variable: $x_{r,t} \geq 0$, the flexible load run in region r at hour t

Objective: $\min_{x, M} \sum_{r,t} x_{r,t} CI_{r,t} (1 - \alpha CFE_{r,t}) + \eta M$

C1
demand met

C2
deadline δ

C3
capacity

C4
geo cap σ

C5
equity η

C6
ramp κ

C7
range ρ

Optimal schedule $x_{r,t}^*$: flexible load in every region and hour

Evaluation (2024–2025)

**Rolling-window
backtest**
731 windows

**Scheduler
comparison**
Oracle, Greedy, FCFS

**Sensitivity
analysis**
one-at-a-time

**Spatio-temporal
decomposition**
routing vs. timing